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A simulation model of bed-occupancy in a critical care unit

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This article focuses on the Critical Care Unit (CCU) of a large teaching hospital. The aim of the study was to optimise the number of beds available in order to minimise cancellations of Elective surgery and maintain an acceptable level of bed-occupancy. The CCU is where critically ill patients are cared for and often requires one-to-one nursing care. The discrete event simulation model, built in Visual Basic for Applications for Excel, seeks to simulate the bed-occupancy of the CCU as well as monitoring any cancellations of Elective surgery. Several 'what-if' scenarios are run including increasing bed numbers, 'ring-fencing' beds for Elective patients, reducing length of stay to account for delayed discharge and changing the scheduling of Elective surgery, and the results are reported.
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1. Introduction

In the UK the National Health Service (NHS) is a very important provision which affects the lives of all residents. Yet despite substantial increases in healthcare expenditure during the past decade (OECD—Health Expenditure, http://www.oecd.org, accessed 17 October 2008), problems such as long waiting times for treatment continue to be reported on a regular basis. It is essential therefore that the NHS works to its optimum capacity in order to meet the ever-increasing demand for healthcare services in the UK.

Health care in the UK is administered through a devolved government department so that each of the constituent countries (England, Northern Ireland, Scotland and Wales) has its own targets and policies. The waiting time target for Wales where this study is based, as set by the National Assembly for Wales, is that by the end of March 2008 no patient should have been waiting for admission as an inpatient or day case for over 22 weeks. In January 2009, there were 20 927 people on the waiting list for inpatient appointment. Of these 54 patients had been waiting for 40 weeks or more with approximately 1.1% of patients already past the 22-week waiting time target (NHS hospital waiting times latest month by patient type (Speciality, Trust, LHB). http://www.statswales.wales.gov.uk, accessed 10 March 2009). The situation is improving but there is still a long way to go. In England, Scotland and Northern Ireland, waiting time is measured from referral to treatment and these targets are 18 weeks, 18 weeks and 15 weeks, respectively (Department of Health, 2008, 18 weeks Referral to Treatment statistics, http://www.dh.gov.uk/ accessed 1, December 2008, New waiting times targets, http://www.scotland.gov.uk/, accessed 10 March 2009 and McGimpsey sets new waiting time targets, 2007, http://www.northernireland.gov.uk/, accessed 10 March 2009).

One factor which can contribute to waiting time problems is the cancellation of Elective surgery. These cancellations occur for many reasons but can generally be classified into one of three groups, namely hospital reasons, clinical reasons or patient reasons. Clinical reasons (worsening of health, for example) and patient reasons (such as patients rethinking their treatment options) are largely out of the control of hospital managers. However, hospital reasons such as insufficient beds available on the Critical Care Unit (CCU) for recovering patients, or that no surgeon is available, can be addressed by managers and can therefore be optimised. This article will focus on one of the hospital reasons, that is ensuring that there are an optimal number of beds available in CCU to minimise cancellations of Elective surgery.

Discrete event simulation has been widely utilised in modelling CCUs, for example Kim *et al* (1999) utilised a simulation model and queueing theory to describe activities in an Intensive Care Unit (ICU) based in Hong Kong. The model was utilised to determine whether the 14-bed capacity of the ICU was sufficient. Also, the reservation of some of the Unit's beds for the sole use of Elective patients was considered. The work undertaken in this study is complementary to this work.

In 2008, Shahani *et al* (2008) utilised Classification and Regression Tree (CART) analysis to create homogeneous groups of patients to feed into a simulation model. Several 'what-if' type scenarios were tested including an increase in

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capacity and the transfer of long-stay patients onto a different ward. CART analysis is a very useful tool for the creation of similar patient groups and has been utilised in many simulation models. Costa *et al* (2003) also used CART analysis to generate similar patient groups in the CCU. Their model differed in the fact that Emergency and Elective patients had the same priority status. This current study assumes that Emergency patients have priority over Elective patients and utilises the arrival sources in the CCU to group patients. Preliminary analysis was undertaken using CART analysis. However, 12 decision variables were required to create homogeneous groups and it was feared that hospital managers would find the model too complicated to be of use.

A dynamic simulation model was built of the CCU at the Cincinnati VA medical Centre, Ohio (Cahill and Redner, 1999). In this model it was stated that the hospital had a high bed-occupancy level (81%). They tested several alternative bed configurations to see whether this level could be reduced to something more acceptable. In the Netherlands, Litvak *et al* (2008) constructed a model of several CCUs in the locality and tested the scenario of reserving a pooled number of beds across the region for emergency admissions. The model could predict the optimal number of regional beds required for any given acceptance rate of emergency admissions. The question of the maximum number of Elective surgeries administered each day to avoid diversion or cancellations of surgeries was addressed by Kolker (2009). The Unit under consideration was large (51 beds), and the optimal number of surgeries scheduled each day was deemed to be 4. A similar piece of analysis follows in this study. Optimal nursing requirements were addressed by Griffiths *et al* (2005). A discrete event simulation model of an ICU was built in Simul8 and various what-if type scenarios relating to nurse numbers were investigated.

This current study utilises discrete event simulation to model patient flow in that setting, by classifying patients according to their arrival source. The aim of the study is to optimise the number of beds required in CCU in order to minimise the cancellation of Elective surgery and yet maintain an acceptable rate of bed-occupancy. Currently, the CCU consists of 24 beds; however, there are five additional beds which may be utilised in times of peak demand. The five additional beds, which are based in the Unit, are only utilised in very exceptional circumstances. These beds are currently unfunded, which means that to use any of them requires the employment of a supplementary nurse due to the 1:1 nurse to patient ratio required in such a ward. These supplementary nurses can cost up to four times as much as a rostered nurse (Griffiths *et al*, 2005) and it is therefore very costly to use any of these additional beds at present. The clinicians at the CCU wanted to ascertain whether any of these five beds should be converted into fully funded CCU beds with the required 1:1 nursing care provision scheduled or whether the Unit is coping with demand by maintaining reasonable bed-occupancy and cancellation levels. The clinicians felt that a tool which could be used to demonstrate to the hospital managers (the key decision makers) the influence of bed numbers on cancellation rates and bed-occupancy levels would be useful.

The main variable, therefore, in the simulation model, was the number of beds. This was incremented from 24 to 29 and results were collected. The main outcome measures were the number of cancellations generated due to a lack of CCU beds, and the mean bed-occupancy level including some insight into the related cost implications. Our main goals were to reduce the number of Elective cancellations while maintaining an appropriate bed-occupancy level in line with the current UK guidelines. The hospital managers have one additional goal—to do the above in a cost-effective manner. The one major constraint which also needed consideration was that after discharge each bed should remain unoccupied for a period of time while the bed and the surrounding area were cleaned intensively. This process can take between 2 and 8 h.

According to the Intensive care society (ICS) (a UK-based society), CCUs should operate at 60–70% capacity (Standards for Intensive Care Units, http://www.ics.ac.uk, accessed 17 October 2008). The European Intensive Care Society gives similar guidance; bed-occupancy should be at around 75% (Accessibility of Intensive Care facilities in Ireland for critically ill patients, www. icmed.com/archive/publications, accessed 10 March 2009). Also, a review of Critical Care facilities from 1985 to 2000 in the USA found that the observed bed-occupancy level for CCUs was 68% (Halpern *et al*, 2004). It would be difficult to say why the guidelines differ from country to country; however, CCUs will always have an optimal capacity level lower than other lower dependency wards to ensure that they have sufficient space to deal with an influx of unplanned admissions where any delay could prove to be disastrous.

1.1. The CCU

The CCU is the sector of the hospital where, as the name suggests, critically ill patients receive treatment. More importantly, a significant proportion of patients who have an operation will be admitted to the CCU for post-operative care. For example, all patients who require ventilation will be cared for in the CCU. This study is based on patients admitted to the CCU in the University Hospital of Wales (UHW) which was formed in 2003.

One of the main factors that differentiates the CCU from other high dependency wards is the level of nursing care needed—the majority of patients in the CCU require one-to-one nursing care. The nurses who care for these patients are specially trained, in short supply and expensive. The department of Health measures their costs for a CCU bed in terms of the number of organs which require support, in other words the number of organ failures. The costs range from £990 for one organ failure to £1741 for five or more organ failures (NHS reference costs 2006–2007, http://www.dh.gov.uk, accessed 18 October 2008). Since the hospital has no control over the number of emergency patients admitted onto the CCU, the focus of this study is Elective patients only. Preliminary data analysis shows that the mean number of organ failures an Elective patient has is 0.36 (SD = 0.73). For this study the cost of an Elective patient in the CCU is assumed to be £990 per day which is the cost of one organ failure. This is the lowest cost published for a CCU patient, using NHS reference costs, and therefore this calculation provides a conservative estimate. This corresponds to a CCU bed costing approximately £361 000 per annum. The model assumes that this is the only cost incurred on the hospital, due to a lack of data referring to overhead costs etc.

1.2. Data

The UHW is the largest hospital in Wales. It has an average of 1064 beds available each day with 86.1% occupied (NHS Beds by specialty (NHS Region, Trust, Hospital, Specialty), http://www.statswales.wales.gov.uk, accessed 17 October 2008). The CCU is also large, with 24 beds currently employed, but with five additional unfunded beds, which may be utilised at times of peak demand.

The data set, which is routinely collected, has complete records between April 2004 and May 2007. It contains a vast amount of information about each patient who is admitted into the CCU including information about arrival type and times, discharge times and physiological factors. In the UHW there is a separate Paediatric CCU, so only data on patients aged 16 or over were used in this analysis. With these restrictions, the data set available for use in the analysis has 4226 patients.

2. Methods

2.1. Summary of the model

A simulation model was built using Visual Basic for Applications for Excel (VBA). VBA was chosen as the tool for building the model since it is run through Excel which is available and familiar to hospital managers and clinicians alike. The model was constructed to allow patients to arrive at the CCU from six different sources, namely Accident and Emergency, the Wards, Elective surgery, Emergency surgery, other hospitals, or the X-Ray department (see Figure 1). Each source had a different inter-arrival time distribution, which was ascertained using the statistical distribution fitting software Stat::Fit utilising the Chi-square goodness of fit test. Statistical distributions were fitted to both inter-arrival times and lengths of stay to ensure that the results were generalisable and that as much variation as possible was attached to these data. Inspection of the length of stay data in particular showed a number of outlying values. These long-stay patients can have a considerable effect on patient flow within the Unit but with a limited number of data points we cannot be certain that all outlying values for length of stay have been accounted for. Therefore statistical distributions, with long tails, were used.

In the model it is assumed that once a patient is booked for admission to the CCU, (s)he is classified as either a ‘planned admission’ (Elective patients only) or as an ‘unplanned admission’ (all other patients). If an arriving patient finds that all beds are occupied, they are sent to a queue. There are two queues built into the model, the ‘Unplanned Admissions’ queue and the ‘Planned Admissions’ queue. The patients in the ‘Planned Admissions’ queue—that is the Elective surgery patients—have their surgery cancelled and are then sent home. The patients in the

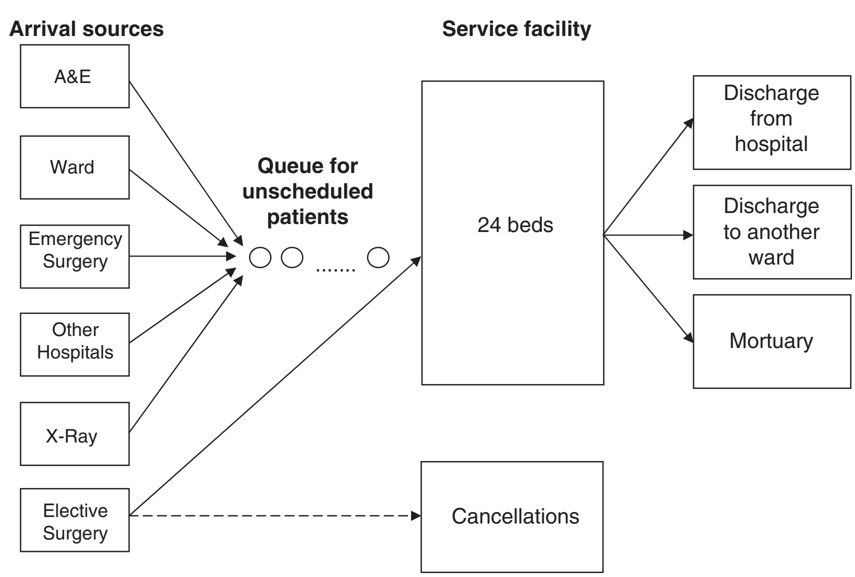

Figure 1 Flow diagram of the CCU simulation model.

'Unplanned Admissions' queue wait until a bed becomes available. This queue wait can have some considerable real-world implications where often alternative arrangements (such as trolley waits) would have to be made. In the model, the patients are then admitted onto the ward and leave the system once their treatment is complete. If a patient arrives at a time where there are unoccupied beds, they are admitted into the CCU and are then treated. The length of stay distributions were linked with the source of arrival.

The model has a user friendly front-end whereby parameters can be changed readily. For example, the mean and standard deviation for each arrival distribution can be changed corresponding to the arrival source. The results from the model are then output to an Excel spreadsheet where they can be analysed further.

2.2. Other calculations

The main aim of this study was to investigate the number of Elective cancellations that could be avoided by having more bed resources, but if a new bed is to be added (at a cost of £990 per day that it is utilised), unplanned admissions will also be able to use it. This is also a worthwhile use of NHS funding as unplanned admissions could be seen more quickly, and trolley waits could be reduced. To calculate this additional benefit approximately, the number of bed days generated by the addition of a bed was calculated and then appropriately scaled by the bed-occupancy rate of the Unit. The number of bed days for Elective and unplanned patients generated from the additional bed was therefore calculated as follows. Using the mean length of stay of Elective patients and the number of additional Elective patients that could be seen given the additional bed, the number of bed days generated by an additional bed was decremented by the number of bed days the Elective patients would use. The average length of stay of unplanned patients was then used to determine the average number of unplanned patients that could utilise the remaining bed days so that this new bed would be full. This crude estimate was a requirement of the clinicians involved in this project and would enable them to have an approximate figure on the use of these additional beds by unplanned admissions.

As was previously mentioned, the model was constructed so that patients would arrive from six different arrival sources and their length of stay would be modelled according to a statistical distribution based upon their arrival source. The following sections describe the inter-arrival time and length of stay distributions for each of the six arrival sources.

2.3. Inter-arrival time distributions

The inter-arrival time distributions were modelled well using the Negative Exponential distribution for five of the six arrival sources (the exception being Elective patients). Table 1 summarises the inter-arrival times for each of the arrival sources. It is noteworthy that the coefficient of variation of the first five sources is approximately 1, which is also consistent with the claim that the distribution is Negative Exponential. This result is not unexpected since arrival distributions for emergency patients are often found to be Negative Exponential (Coats and Michalis, 2001; Moore, 2003) due to the random nature of emergency events.

The inter-arrival time of Elective patients, over the period of a week, is time-dependent in nature (see Figure 2). This graph shows that for many arrivals the inter-arrival time is between 0 and 4 h. Another peak occurs between 22 and 26 h, and then other smaller peaks at approximately 24 h intervals. These observations are consistent with the practice of Elective surgery being performed at specific sessions spaced at 24 h intervals.

Inspection of the hour at which each Elective patient arrives at the Unit showed that the majority of arrivals occur in the early evening with very few in the early hours of the morning. Because of this time-dependency, it is not a trivial task to model the arrival times of these patients. It was also evident from the data that if a day is defined to begin and end at 07:00, the arrival times of Elective surgery patients could be modelled using the Normal distribution. The fitted Normal distribution had a mean value of 17.91 and a standard deviation of 3.16. Unfortunately, the fit is not statistically significant, but it gives a good indication of the pattern of arrival times.

2.4. Length of stay distributions

There are six different length of stay distributions, each of which relates to a specific arrival source. This may not be

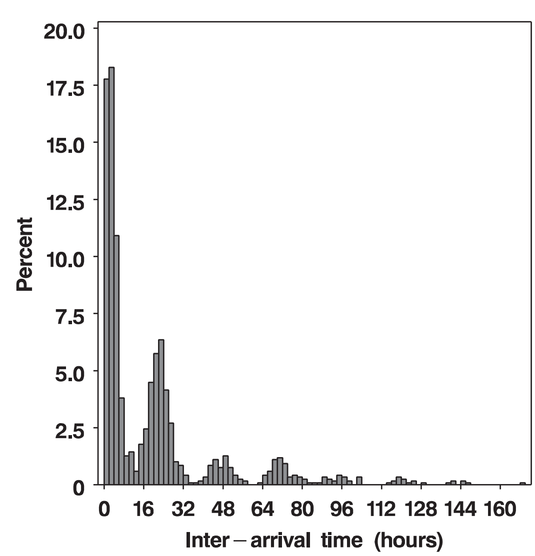

Figure 2 Elective surgery inter-arrival times.

intuitive but the data indicate significant differences between the length of stay distributions recorded in hours for each of the arrival sources ($P<0.0001$).

Chi-square goodness of fit tests were performed using Stat::Fit and it was found that the lognormal distribution was a good fit for each of the groups of patients, other than Elective surgery. The lognormal distribution is commonly fitted to length of stay data (Lowery, 1993; Litvak *et al*, 2008) due to its long tails. Also, Hyperexponential distributions were fitted to the data to ensure that a phase-type distribution would not improve results. Up to five phases were used. Phase-type distributions have been used widely to fit length of stay data, for example Marshall and McClean (2004). Table 1 shows the theoretical and actual values for the mean and standard deviations of the length of stay as well as the distribution finally used. For most of the groups, the observed values compare favourably with the theoretical ones.

Once again, the length of stay distribution of Elective patients has a similar pattern to the inter-arrival distribution. This is a result of the time of day at which the clinician make their ward rounds and patients cannot be referred for discharge until the ward round has been completed. Rather than trying to find a distribution which adequately models this data, the lengths of stay of patients arriving from Elective surgery were sampled directly from the data.

After consultation with hospital managers and clinicians at UHW, it was clear that the actual length of stay of a patient was not a true reflection of the amount of time a bed was used. After each discharge, the bed and the surrounding area require intensive cleaning. This can take anything between 2 and 8 h. Since no data are collected on this issue, it was assumed that a changeover time of 5 h was required after discharge. A sensitivity analysis of this value followed, changing it from 5 h down to 2 and up to 8 and the resulting bed-occupancy levels ranged from 19.8 to 20.4, respectively. These figures are relatively close to the mean value quoted in this article (20.1) showing that the simulation model is not very sensitive to the cleaning time parameter.

2.5. Verification and validation

To allow the model to enter steady-state conditions, a warm-up period of 1 month was included. The model was subsequently run for 1 year and then replicated 100 000 times. Validation of the model consisted of comparing the actual data with the output from the model. The model was run with 24 beds available—that is, the number of beds currently fully funded in the CCU. This resulted in 57 cancellations per year and an 82% bed-occupancy rate, compared with observed rates of cancellation and bed-occupancy of 57 and 87%, respectively. The raw data only include patients who have actually entered the CCU and hence had not had their operations cancelled. By looking at the data, it was clear that on occasions more than the 24 funded beds were occupied in the CCU at any one time and

Table 1 Summary measures for inter-arrival times and lengths of stay for patients from various sources

Source	*N*	*Distribution*	$\chi^2(v)$, *p*-value	*Theoretical mean*	*Theoretical standard deviation*	*Actual mean*	*Actual standard deviation*
Inter-arrival distribution (in hours)							
A&E	1134	Negative Exponential	53 (19), $p<0.01$	22.72	22.72	22.72	23.24
Ward	916	Negative Exponential	34 (17), $p<0.01$	26.0	26.0	25.97	26.59
Emergency	715	Negative Exponential	32 (15), $p<0.01$	37.0	37.0	37.03	38.95
Other hospital	235	Negative Exponential	14 (7), $p=0.05$	47.2	47.2	47.18	49.49
X-ray	44	Negative Exponential	10 (4), $p=0.04$	575.0	575.0	575.09	630.64
Elective	1182	—	—	—	—	19.51	26.42
Length of stay distribution (in hours)							
A&E	1133	Hyperexponential (4-stage)	—	128.79	267.51	123.59	304.40
Ward	907	Hyperexponential (4-stage)	—	177.89	276.54	181.73	273.61
Emergency	711	Hyperexponential (3-stage)	—	140.15	218.02	141.02	218.07
Other hospital	233	Lognormal	31 (19), $p=0.04$	212.86	457.67	210.13	431.48
X-Ray	44	Lognormal	16 (7), $p=0.03$	87.53	108.15	91.27	124.33
Elective	1181	—	—	57.34	99.78	—	—

the additional unfunded beds were used. In fact, 29 beds were occupied at one stage. Indeed, initial analysis revealed that on 9% of occasions there were more than 24 beds occupied; that is, one or more of the extra five unfunded beds stored for peak demand use were utilised. Taking account of the length of stay of patients, this amounts to over 7% of the available CCU time. After consultation with staff, it was suggested that when a patient dies it takes time for the nurses to prepare that bed for the next patient which means that patients would feasibly be waiting on a trolley for a period of time. In order to replicate this situation in validating the simulation model, it was necessary to increase the number of beds available. It is stressed that this was only done to help the validation procedure and to produce results which were comparable with the raw data.

The number of admissions in an average year, according to the data, was 1359, and the simulation recorded 1341 admissions (when sufficient beds were used to satisfy demand). The mean number of beds occupied at any time during the year was 20.10 according to the data, compared with 20.23 according to the simulation. This difference was not found to be significant at the 95% level.

3. Results and discussion

The model was run for several 'what-if' scenarios including changing bed-numbers, ring-fencing beds for Elective and Emergency demand, eradicating bed-blocking and changing surgery scheduling. Each will be discussed separately.

3.1. Changing number of funded beds

Firstly, the number of beds in the Unit was incremented from 22 (which is fewer than the number currently funded in the Unit) to 29 (which is the maximum number which could be accommodated if all extra beds were fully funded) and the results are displayed in Table 2.

It is clear that the number of cancellations and bed-occupancy levels drop as additional beds are provided. Table 2 also summarises the incremental cost of each cancellation avoided. As the number of additional beds increases, the cost of each avoided cancellation increases. This dramatic increase is due to the fact that as beds are added, fewer cancelations are avoided, and since the cost of the occupied bed remains constant, the cost per avoided cancellation will be higher. The current formation of the Unit, with 24 funded beds, is taken to be the baseline. The five additional unfunded beds were not included in the baseline figures since they are very costly to run at present due to the supplementary nursing requirement. This formation resulted in 57 Elective cancellations with a bed-occupancy level of 82%. This cancellation level translates to roughly five cancellations per month which is very high considering that there are only approximately 30 Elective admissions scheduled each month. Also, the bed-occupancy level of 82% is considerably higher than the 60–70% occupancy guidelines given by the ICS. With 25 beds in the CCU, the model showed that the number of cancellations would drop to approximately 20 per year with 80% bed-occupancy. The cost of the extra bed is around £361 000 per annum (assuming only one organ failure per Elective patient); therefore the resulting reduction of 37 cancellations implies that the cost per avoided cancellation is £8200 (see Table 2). Twenty nine beds, which is the maximum number which could be accommodated, results in no cancellations and a bed-occupancy level of 70%. Clearly, the bed-occupancy level of 70% is desirable and meets the ICS guidelines. The important question to consider is whether adherence to the ICS guidelines warrants the extra investment required for the additional CCU beds. This is a matter for stakeholders to discuss. Turning to the issue of cancellations, with only 27 beds no Elective cancellations occur implying that the investment in these extra beds, in terms of Elective cancellations avoided, is wasted.

3.2. Ring-fencing beds for Emergency and Elective demand

Secondly, we considered 'ring-fencing' a number of beds to be reserved for unplanned admissions and reserving the remainder for admissions from Elective surgery.

Table 2 Summary measures using simulation model and costing

Number of beds	*Mean number of occupied beds*	*% occupied*	*Number of cancellations*	*Number of cancellations avoided*	*Cost per avoided cancellation*	*Number of additional emergency patients admitted*
22	19.15	87	146	0	—	0
23	19.40	84	101	45	£8000	2
24	19.76	82	57	89	£8200	6
25	20.05	80	20	126	£9800	14
26	20.18	78	3	143	£21 200	26
27	20.20	75	0	146	£120 300	38
28	20.20	72	0	146	—	50
29	20.20	70	0	146	—	62

To achieve the 60–70% target recommended by the ICS in both the Elective and Emergency beds, 25 Emergency beds would be required and 1 Elective bed. This would result in 295 cancellations each year. Clearly this is not an acceptable number of cancellations (the 60–70% bed-occupancy target would also be met with no Elective beds and 25 Emergency beds, if cancellations were not taken into account, but this configuration would not be considered since it is clinically unacceptable). If four Elective beds were funded, no Elective cancellations would occur, but these beds would only be utilised 65% of the time. Comparison of this 'ring-fenced beds' scenario with the CCUs current formation indicates that to achieve the ICS bed-occupancy target and to achieve a low number of cancellations, 29 beds are required, thus costing in the very least an additional £1.8 million (assuming that all patients who utilise these beds have one organ failure), which suggests that 'ring-fencing' is not beneficial.

3.3. Eradicating bed-blocking

Thirdly, some additional data were obtained referring to the delay experienced by a patient prior to discharge, but only for the period of time from 1st of April 2004 to 31st of December 2005. From these data it was evident that 57% of patients who were admitted onto the CCU during this time period experienced a delayed discharge, and the mean delay experienced by a patient was 33.64 h. Investigation into the delay data found considerable time dependencies; therefore to account for this delay, a delay value (ranging from 0 to 320 h) was sampled from the original delayed discharge data and this value was subtracted from the calculated length of stay. Again, the model with this decreased length of stay was run for several different bed numbers. The results can be found in Figure 3.

Figure 3 demonstrates that the eradication of bed-blocking will cause a reduction in Elective cancellations.

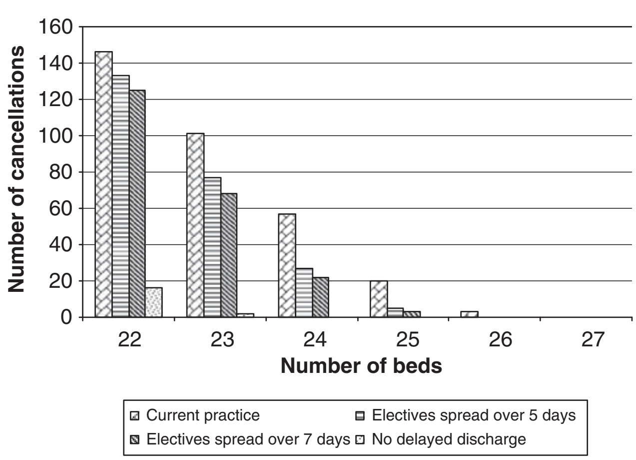

Figure 3 Number of Elective cancellations per year, for current practice, Electives spread over 5 days, Electives spread over 7 days and removing bed-blocking.

In fact, if bed-blocking could be eradicated, with 24 beds no Elective cancellations would occur. The current formation of 24 beds yielded a 72% bed-occupancy level also. The addition of one bed causes the bed-occupancy rate to drop to 69% thus complying with the ICS guideline. The cost implications for reducing delayed discharge are unknown, since there are many reasons why a patient could block a CCU bed, ranging from the simple (ie too few porters, too few pharmacists) to the very complex (no available beds in the discharge ward). Further investigation into the reasons for this delayed discharge is required before any definite cost analysis can take place.

3.4. Changing the scheduling of surgery

Finally, it was clear from the CCU admissions data that the majority of Elective surgery (around 83%) is performed between Tuesday and Friday. The remaining 17% is mostly on Monday, with 3% spread over the weekend. It is of interest to discover whether a more uniform distribution of Elective admissions is beneficial to the overall running of the Unit given that the average number of Elective surgeries is approximately 365 per year. Two final scenarios, relating to the scheduling of Elective surgery were considered. The first, being that Elective surgery would take place daily, with one operation each day (for seven days). For this scenario to be implemented, a significant change to working practice of surgeons would need to be agreed. The second, being that either one or two surgeries are performed each day during the five days, Monday to Friday. The number of cancellations which occur annually are shown in Figure 3. Bed-occupancy levels do not vary much given the modification of surgery times (to scheduling over 5 days a week, or 7 days a week) and this is to be expected; however, the reduction in cancellation rates is substantial for both scenarios. For example, when Elective surgery is performed over 5 days with either one or two surgeries performed daily, with 24 beds only 27 cancellations occur (compared with the current practice causing 57 cancellations). If Elective surgery took place over 7 days, with 24 beds a larger affect once again is seen with only 22 cancellations occurring. It is therefore clearly beneficial to schedule the Elective surgery over at least 5 days and up to 7 days but the latter would require major changes in surgeons' working conditions which may result in clinician opposition. The cost implication of these changes in surgeons working practice, in particular the rescheduling of surgery onto 5 days, are minimal. The operating theatres are set up daily for surgery since Emergency patients use them as well as Elective surgeries that do not require CCU after care. However, these other Elective surgeries could be impacted; for example, very often many patients who require routine surgery will all have their surgery within a narrow time slot. This rescheduling of CCU Elective patients could affect these others.

4. Conclusion

The CCU is only able to achieve the high level of bed-occupancy by utilisation of the five additional beds when circumstances demanded this extra capacity. If the main 24 beds alone were used, the bed-occupancy would have been 82%. The CCU, under its current formation, does not adhere to ICS bed-occupancy guidelines. Also, the waiting lists associated with Elective surgery patients who require a period of care in the CCU will also increase due to the number of cancellations which occur at the Unit. In order to meet the ICS guidelines five additional CCU beds are required. With these additional beds, no cancellations of Elective surgery will occur. This is a simple solution yet requires substantial investment.

This study has highlighted several different configurations of the management of the CCU including resolving the issue of bed-blocking, 'ring-fencing' beds for Elective and unplanned admissions and changing the working practice of theatre staff. The most beneficial action for reducing bed-occupancy levels and cancellation rates is to resolve the issue of bed-blocking. If this were possible, the CCU would only require one additional bed to meet the ICS bed-occupancy guidelines and no cancellations would occur, however the cost implications are unknown.

If this were not possible, the next best option is to change the working practice of theatre staff. If Elective surgery was performed daily, the number of annual cancellations would drop considerably. This option would require the backing of clinicians and may be difficult to administer but with the compelling evidence documented in this research the clinicians may be persuaded. However, if Elective surgery was performed on 5 days per week, in a more regular fashion, cancellation rates would also drop. This second option is much simpler to implement, with minimal disruption to theatre staff.

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